

Drone Designer

Project Summary & Architecture Guide

A **VB.NET WinForms** desktop application (.NET Framework 4.7.2) that takes UAV mission requirements as input and recommends optimal drone hardware components (motors, batteries, ESCs, flight controllers, cameras, GPS, and more).

The application implements a **three-stage selection pipeline** (Module 1) and a **SolidWorks CAD generation pipeline** (Module 2), backed by a JSON component database of real-world drone hardware.

1. Project File Tree

```
Drone Designer/  
  Drone Designer.slnx  
  Drone Designer.vbproj  
  App.config  
  packages.config  
  
  Core/  
    Data/  
      ComponentRepository.vb  
    Interfaces/  
      IComponentSelector.vb  
    Models/  
      ComponentSpecs.vb  
      MissionSpecs.vb  
      ComponentDisplayRow.vb  
      PipelineResult.vb  
    Services/  
      ComponentSelectionEngine.vb  
      PipelineOrchestrator.vb  
  
  UI/  
    Forms/  
      MainForm.vb  
      MainForm.Logic.vb  
      MainForm.CAD.vb  
      MainForm.Designer.vb  
      CadProgressForm.vb  
    Controls/  
  
  Solidworks/  
    SolidWorksAutomation.vb  
    MacroRunner.vb  
    Module2UsageExample.vb  
  
  Utilities/  
    ConfigManager.vb  
  
  Resources/  
    components.json  
    Macros/  
      MotorMount.swb  
  
  packages/  
    Newtonsoft.Json.13.0.4/
```


3. File-by-File Breakdown

ComponentSpecs.vb

Core/Models/ComponentSpecs.vb | ~1,314 lines

Purpose: Data model backbone. Defines 11 typed component subclasses and all supporting enumerations.

- Abstract base class **ComponentSpecs** with universal properties: Id, Name, Manufacturer, MassGrams, Dimensions, Voltage, Temperature range, Price.
- Concrete subclasses: **MotorSpec**, **ESCSpec**, **PropellerSpec**, **FlightControllerSpec**, **BatterySpec**, **GPSSModuleSpec**, **TelemetryRadioSpec**, **CameraSpec**, **ServoSpec**, **ReceiverSpec**, **PowerDistributionBoardSpec**.
- Key enums: ComponentCategory, MotorType, BatteryCellChemistry, GNSSConstellation, ESCProtocol, FlightControllerFirmware, CameraOutputInterface, PowerConnectorType.
- Computed read-only properties (e.g., BatterySpec.NominalPackVoltageV, MaxContinuousCurrentAmps).

Connections: Consumed by ComponentRepository (deserialization target), ComponentSelectionEngine (selection logic), ComponentDisplayRow (display mapping), and MacroRunner (parameter extraction).

MissionSpecs.vb

Core/Models/MissionSpecs.vb | ~458 lines

Purpose: Complete set of UAV mission parameters. Primary input to the selection engine.

- Flight performance: EnduranceMinutes, MaxRangeKm, CruiseSpeedMs, MaxAltitudeMeters, MaxWindSpeedMs, MaxTakeoffMassGrams.
- Payload: PayloadMassGrams, PayloadDimensions (L x W x H).
- Environment: MinOperatingTempCelsius, MaxOperatingTempCelsius, RequiresWaterproofing, RequiredIPRating.
- Mission profile & config: MissionProfileCategory, UAVConfiguration, PowerSourceType, RegulatoryClass.
- Dual enum sets for backward compatibility (legacy + engine-facing enums).

Connections: Built by MainForm.Logic.BuildMissionSpecs(). Passed to ComponentSelectionEngine and PipelineOrchestrator.

PipelineResult.vb

Core/Models/PipelineResult.vb | ~368 lines

Purpose: Immutable result types for the end-to-end pipeline (Task 15).

- **PipelineStage** enum: Validating, SelectingComponents, ConnectingToSolidWorks, GeneratingPart, SavingFile, WritingManifest, Finalising, Failed.
- **PipelineProgressReport**: immutable snapshot with Stage, PercentComplete, StatusMessage, DetailMessage, ErrorMessage. Used via IProgress(Of T).
- **GeneratedPartRecord**: describes a single CAD file produced (ComponentId, PartType, FilePath, Success/ErrorDetail).
- **PipelineResult**: final outcome with Success flag, GeneratedParts list, OutputDirectory, ManifestPath, Duration, ToSummaryString() for display.

Connections: Produced by PipelineOrchestrator. Consumed by MainForm.CAD.vb and CadProgressForm.

ComponentSelectionEngine.vb

Core/Services/ComponentSelectionEngine.vb | ~1,544 lines

Purpose: Core selection algorithm. Implements the three-stage pipeline (Tasks 7, 8, 9).

- **Task 7 (Weight & Thrust):** EstimateMtow() uses fixed-point iteration. SelectMotors() and SelectPropellers() filter and rank candidates.
- **Task 8 (Power System):** CalculatePowerBudget() determines cell count, currents, capacity. SelectBatteries(), SelectEscs(), SelectPdb().
- **Task 9 (Avionics):** SelectFlightController(), SelectGpsModule(), SelectTelemetryRadio(), SelectReceiver(), SelectCamera().
- Returns **SelectionResult** with up to 5 candidates per category plus intermediate sizing data.

Connections: Implements IComponentSelector. Reads from ComponentRepository. Called by MainForm.Logic and PipelineOrchestrator.

PipelineOrchestrator.vb

Core/Services/PipelineOrchestrator.vb | ~559 lines

Purpose: Coordinates the full end-to-end workflow: MissionSpecs -> Selection -> SolidWorks -> CAD parts (Task 15).

- RunAsync(): 6-stage async pipeline: Validate, Select Components, Connect SW, Generate Parts, Write Manifest, Finalise.
- RunFromSelectionAsync(): skips re-running selection when user already has a SelectionResult from Module 1.
- Partial-failure tolerant: if one motor mount fails, remaining motors are still attempted.
- CancellationToken supported: UI Cancel button hooks in. IProgress for all status updates.
- Writes a plain-text component_manifest.txt alongside generated CAD files.

Connections: Consumes IComponentSelector, SolidWorksAutomation, MacroRunner, ConfigManager. Called by MainForm.CAD.vb.

ComponentRepository.vb

Core/Data/ComponentRepository.vb | ~304 lines

Purpose: Read-only data access layer. Loads and indexes components from JSON.

- Loads Resources/components.json on construction; builds category and ID indexes.
- Typed accessors: GetMotors(), GetBatteries(), GetESCs(), GetFlightControllers(), etc.
- Filter methods: GetMotorsWithMinThrust(), GetBatteriesWithMinCapacity(), GetComponentsUnderMass().
- Uses Newtonsoft.Json for flexible deserialization. Warns on duplicate IDs without crashing.

Connections: Reads components.json. Injected into ComponentSelectionEngine constructor.

IComponentSelector.vb

Core/Interfaces/IComponentSelector.vb | ~31 lines

Purpose: Contract interface for the selection engine.

- Single method: SelectComponents(specs As MissionSpecs) As SelectionResult.
- Enables dependency injection, mocking for tests, and future web-service migration.

Connections: Implemented by ComponentSelectionEngine. Used by PipelineOrchestrator and MainForm.Logic.

ComponentDisplayRow.vb

Core/Models/ComponentDisplayRow.vb | ~236 lines

Purpose: Display-only DTO bridging ComponentSpecs properties to DataGridView column names.

- Factory method FromComponentSpecs() creates display rows from engine output.
- Helpers: FormatTempRange(), BuildDimensionsString() for human-readable formatting.

Connections: Created by MainForm.Logic.DisplaySelectionResult(). Bound to DataGridView in MainForm.

MainForm.vb

UI/Forms/MainForm.vb | ~1,023 lines

Purpose: Root application window. Two-tab layout built entirely in code.

- Tab 1 (Input): GroupBoxes for Flight, Payload, Environment, Mission parameters.
- Tab 2 (Output): DataGridView for results + summary panel + Generate CAD button.
- All controls created programmatically. Form size: 1180 x 760.

Connections: Partial class with MainForm.Logic.vb, MainForm.CAD.vb, and MainForm.Designer.vb.

MainForm.Logic.vb

UI/Forms/MainForm.Logic.vb | ~492 lines

Purpose: Event handlers, validation, and Module 1 engine wiring.

- OnLoad(): Initializes ComponentRepository and ComponentSelectionEngine.
- ValidateFormInputs(), BuildMissionSpecs(), DisplaySelectionResult().
- Enum mapping helpers: ParseMotorCount(), MapMissionProfileType(), MapEnvironmentType().

Connections: Calls ComponentSelectionEngine.SelectComponents(). Updates DataGridView.

MainForm.CAD.vb

UI/Forms/MainForm.CAD.vb | ~410 lines

Purpose: Partial class for MainForm - Module 2 (SolidWorks CAD) button wiring (Task 15).

- WireCADControls(): hooks the Generate CAD button click handler.
- btnGenerateCAD_Click(): validates selection exists, picks output folder, runs PipelineOrchestrator async.
- Shows CadProgressForm (modeless) with real-time progress. Offers to open output folder on success.
- Lazy-initialises SolidWorksAutomation and PipelineOrchestrator on first use.

Connections: Uses PipelineOrchestrator.RunFromSelectionAsync(). Shows CadProgressForm. Reads _lastSelectionResult from Logic.

MainForm.Designer.vb

UI/Forms/MainForm.Designer.vb | ~31 lines

Purpose: Auto-generated form designer stub.

Connections: Partial class with MainForm.vb.

CadProgressForm.vb

UI/Forms/CadProgressForm.vb | ~397 lines

Purpose: Modal progress dialog shown while the SolidWorks pipeline runs (Task 15).

- Dark-themed form with stage label, progress bar, detail label, scrolling activity log.
- Cancel button signals CancellationTokenSource; swaps to Close button on completion.
- Exposes IProgress(Of PipelineProgressReport) for the orchestrator to report into.
- PipelineCompleted event raised when user dismisses the dialog.
- Thread-safe: all public methods use Me.Invoke for cross-thread marshalling.

Connections: Created by MainForm.CAD.vb. Receives progress from PipelineOrchestrator. Returns PipelineResult via event.

SolidWorksAutomation.vb

Solidworks/SolidWorksAutomation.vb | ~547 lines

Purpose: Manages the COM connection lifecycle to SolidWorks (Task 12/14).

- Connect(): tries GetActiveObject first (attach), then CreateObject (launch new instance).
- Disconnect(): closes SW only if we launched it and no docs are open.
- IsConnected(): liveness ping via COM round-trip. GetVersion(): maps API revision to year.
- Late binding (Object) so the project compiles without SolidWorks installed.
- IDisposable for guaranteed cleanup. COMErrorCodes module for friendly HRESULT messages.

Connections: Used by PipelineOrchestrator and MainForm.CAD.vb. Provides Application property to MacroRunner.

MacroRunner.vb

Solidworks/MacroRunner.vb | ~500+ lines

Purpose: Bridge between ComponentSpecs and SolidWorks macro execution (Task 13).

- MacroParameters class: flat key-value store with FromComponentSpec() factory.
- RunMacroOnTemplate(): opens template part, injects parameters as custom properties, runs macro via RunMacro2, saves output.
- Parameter passing via custom document properties + temp INI file fallback.
- Cleans up temporary properties and files after macro completion.

Connections: Uses SolidWorksAutomation.Application COM object. Called by PipelineOrchestrator. Reads ComponentSpecs.

Module2UsageExample.vb

Solidworks/Module2UsageExample.vb | ~110 lines

Purpose: Reference snippet (not compiled) showing how to wire SolidWorks automation from the UI.

- Documents expected MechanicalSpecs keys for motor components (OuterDiameter, BoltCircleDiameter, etc.).
- Includes deployment checklist for SolidWorks + macro setup.

Connections: Documentation only. To be deleted once actual UI integration is complete.

MotorMount.swb

Resources/Macros/MotorMount.swb | ~399 lines

Purpose: SolidWorks VBA macro that generates a parametric motor-mount part.

- Reads parameters from %TEMP%\SW_MotorMountParams.ini (KEY=VALUE format).
- Builds geometry: base disc extrusion, central shaft hole, motor bolt holes on bolt circle, arm attachment holes.
- Stamps custom properties (DD_* prefix) on the part for traceability.
- Writes status to %TEMP%\SW_MacroStatus.ini for the VB.NET host to poll.

Connections: Executed by MacroRunner via swApp.RunMacro2. Parameters written by SolidWorksAutomation.

ConfigManager.vb

Utilities/ConfigManager.vb | ~313 lines

Purpose: Application-wide configuration manager (singleton pattern).

- Loads/saves DroneDesigner.config.json (auto-creates with defaults if missing).
- AppSettings: ComponentsDatabasePath, SolidWorksInstallPath, TemplatePartsDirectory, OutputDirectory, LogFilePath, LogLevel.
- Resolved properties convert relative paths to absolute. Uses DataContractJsonSerializer.

Connections: Used by ComponentRepository, PipelineOrchestrator, and MainForm.CAD.vb.

components.json

Resources/components.json | ~1,829 lines

Purpose: Component database. JSON array of real-world drone hardware.

- Categories: motor, esc, propeller, flight_controller, battery, gps, telemetry, camera, servo, receiver, pdb.
- Includes real hardware: T-Motor, Pixhawk, u-blox, EMAX, Matek, etc.

Connections: Loaded by ComponentRepository.vb on startup.

packages.config

packages.config | ~4 lines

Purpose: NuGet package manifest. Lists Newtonsoft.Json 13.0.4 targeting net472.

Connections: Consumed by NuGet restore. Newtonsoft.Json used by ComponentRepository for JSON deserialization.

App.config

App.config | ~6 lines

Purpose: .NET Framework startup configuration. Targets .NET 4.7.2.

Connections: Read by .NET runtime at startup.

Drone Designer.vbproj

Drone Designer.vbproj | ~135 lines

Purpose: MSBuild project file. Defines compilation targets, references, and resources.

- Target: .NET Framework 4.7.2, WinExe output.
- References: System, System.Drawing, System.Windows.Forms, Newtonsoft.Json.

Connections: Defines the build for the entire project.

4. Summary Table

File	Lines	Role
ComponentSpecs.vb	1,314	11 component subclasses + enums
MissionSpecs.vb	458	Mission input parameters
PipelineResult.vb	368	Pipeline progress/result types (Task 15)
ComponentSelectionEngine.vb	1,544	3-stage selection pipeline (Module 1)
PipelineOrchestrator.vb	559	End-to-end pipeline coordinator (Task 15)
ComponentRepository.vb	304	JSON data access & indexing
IComponentSelector.vb	31	Engine interface contract
ComponentDisplayRow.vb	236	Display DTO for DataGridView
MainForm.vb	1,023	Root window, two-tab layout
MainForm.Logic.vb	492	Module 1 validation, wiring, display
MainForm.CAD.vb	410	Module 2 (Generate CAD) button wiring
MainForm.Designer.vb	31	Auto-generated stub
CadProgressForm.vb	397	CAD generation progress dialog
SolidWorksAutomation.vb	547	SolidWorks COM connection manager
MacroRunner.vb	500+	Macro parameter injection & execution
Module2UsageExample.vb	110	Reference snippet (not compiled)
MotorMount.swb	399	VBA macro: parametric motor mount geometry
ConfigManager.vb	313	App config & path resolution
components.json	1,829	Hardware database (JSON)
packages.config	4	NuGet manifest (Newtonsoft.Json)
App.config	6	.NET Framework 4.7.2 startup
Drone Designer.vbproj	135	MSBuild project definition

5. Key Design Patterns

- **Repository Pattern** - ComponentRepository abstracts the JSON data source behind typed query methods.
- **Dependency Injection** - ComponentSelectionEngine and PipelineOrchestrator accept dependencies in constructors.
- **Interface Segregation** - IComponentSelector decouples UI from the concrete engine implementation.
- **Enum-Over-Strings** - ComponentCategory, MotorType, BatteryCellChemistry, etc. prevent magic strings throughout.
- **Immutable Intermediate Types** - MtowEstimate, PowerBudget, PipelineProgressReport expose transparent sizing breakdown.
- **Partial Classes** - MainForm split across .vb (layout), .Logic.vb (Module 1), .CAD.vb (Module 2), .Designer.vb (scaffold).
- **Async Pipeline** - PipelineOrchestrator uses Async/Await + IProgress(Of T) for non-blocking UI during CAD generation.
- **Late Binding** - SolidWorksAutomation uses Object (not early-bound COM types) so the project compiles without SolidWorks installed.

6. Extension Points

- **Additional CAD Parts** - New .swb macros can be added to Resources/Macros/ and wired into PipelineOrchestrator for propeller mounts, battery trays, etc.
- **Module 3 (Cost / Supply Chain)** - SelectionResult exposes up to 5 candidates per category for trade-off analysis.
- **Web Migration** - IComponentSelector interface and plain data types enable a REST API without UI changes.